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G2 Amiga release 1.0 · 2026-11-03

USB Host of the G2 Amiga

The G2's four USB sockets behind its USB2514 hub, run by an EHCI controller in the fabric and the RISC-V firmware: the stick that holds every disk image, the keyboard, the mouse, two game pads and one serial or MIDI device, all hot-pluggable.

HIGHLIGHTS

- EHCI 1.0 controller (Xilinx xps_usb_host v1.01.a) on a USB3300 ULPI PHY.
- USB2514 hub: four sockets; full-speed devices through its transaction translator.
- Mass storage, HID boot keyboard and mouse, DualShock 3/4, USB-MIDI, FTDI serial and MPSSE.
- 32 KB DMA RAM, double-buffered bulk transfers: 13.14 MB/s stick reads.
- Hot plug on every port, scanned every 2 s.

KEY PARAMETERS

PARAMETER	VALUE
Controller	EHCI 1.0, 1 root port
PHY	USB3300, ULPI
Hub	USB2514, 4 ports
Speeds	high, full, low
DMA RAM	32 KB
Bulk read	13.14 MB/s
Interrupt slots	6

1 STRUCTURE

BLOCK	PART	FUNCTION
Host controller	Xilinx XPS USB Host Controller xps_usb_host v1.01.a (DS734), EHCI 1.0, one root port	High speed; full and low speed through split transactions; licensed EDK core, encrypted netlist
PHY	SMSC USB3300, ULPI 8-bit SDR at 60 MHz	The line interface; its 24.000 MHz reference is made by a fabric PLL (125 MHz × 24 / 5 / 25)
Hub	Microchip USB2514, 4 downstream ports, ganged power	Every external USB socket of the G2; its transaction translator carries full-speed devices
Bridge	Wishbone to PLB, 62.5 MHz PLB clock	Register window at 0x8100_0000: local registers, EHCI registers at +0x100
DMA RAM	32 KB of dual-port block RAM at 0x8101_0000	Every descriptor and data buffer the controller reads or writes; the CPU copies data in and out
Driver	RISC-V firmware: EHCI schedule, hub, enumeration, class drivers	Control, bulk and interrupt transfers; no isochronous use in this release

2 MEMORY AND TRANSFERS

ITEM	VALUE
Descriptor byte order	EHCI descriptors are big-endian words to the core: the firmware swaps every descriptor word
Control	Setup at DMA RAM 0x0E00, data buffer 448 bytes at 0x0E40
Bulk	Two ping-pong buffers of 12 KB (0x1000–0x6FFF) and two qTDs: chunk k + 2 is queued into the buffer chunk k has left while the CPU copies the finished one
Interrupt	Queue heads in the periodic schedule, one slot per device endpoint: 1 keyboard, 2 mouse, 3 and 4 pads, 5 and 6 the MIDI or FTDI device (IN with interrupt on completion, OUT)
Bulk rate	13.14 MB/s in a 16 MB stick read benchmark; 10,473 KB/s for a 600,604-byte file (measured on the unit's shell firmware)
Limit	The CPU copy out of the uncached DMA RAM

3 DEVICE CLASSES

CLASS	DRIVER	USE IN THE AMIGA FIRMWARE
Mass storage, bulk-only transport, SCSI	Block read and write, any sector size up to 4 KB	The stick: FAT12/16/32 and exFAT (FatFs R0.16, long names, code page 437)
HID boot keyboard and mouse	SET_PROTOCOL boot, SET_IDLE, interrupt reports	Amiga keyboard and mouse
Sony DualShock 3 / 4	Start sequences of each pad, 64-byte reports	Two Amiga joysticks
USB-MIDI 1.0 (class compliant)	Event packets, cable 0	Paula's serial port at the MIDI rate
FTDI FT232R, FT2232H and compatible	Vendor requests, 2-byte status per packet, latency timer; MPSSE on an FT2232H's channel A	Paula's serial port, the serial FIFO board, or an MPSSE byte stream
Hub	Port status, change pipe	The on-board USB2514; external hubs not supported

4 HUB PORTS

BEHAVIOUR	DETAIL
Scan	Every 2 s each port's status is read; a connection change enumerates or releases the device
Power	Set once at start and never cleared: clearing PORT_POWER on the ganged hub removes every device
Stick	The first mass-storage device found holds \amiga; its port is skipped by the scan
Failures	A device without a device descriptor after 3 rounds is rejected and retried after 30 s; the reason is in BOOT.LOG

5 LIMITS AND UNTESTED ITEMS

ITEM	STATE
Low speed (1.5 Mbit/s)	Supported: the firmware's USB driver runs low-speed devices as split transactions through the hub's transaction translator (DS734 lists low speed as unsupported by the core alone). Low-speed keyboards and mice tested on hardware
Isochronous (audio, video class)	Not used
External hubs	Not supported
Licence	The controller is a fee-based Xilinx EDK core: its netlist and the firmware USB drivers are not part of the public source

Xilinx DS734 v1.01a; SMSC USB3300 and Microchip USB2514 data sheets; measurements on a G2 revision C unit. Repository mikewolak/pano-g2-amiga, release 1.0.